

Audio Analysis Made Easy Using Essentia

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Essentia is an open source C++ library for audio analysis, released under Affero GPL V3 licence. The package includes a complete set of algorithms required for extracting information from audio, audio file input/output functionality, standard digital signal processing (DSP) building blocks, filters, generic algorithms for statistical characterisation, and spectral, temporal, and high-level music descriptors.

Researchers make use of the Essentia environment for experimentation and rapid application development. For analysing a large database of audio tracks, the software enables compilation of optimised extractors to run efficiently on computing clusters. Users who are familiar with MATLAB/Python environments can make use of Python bindings in the package. The latest release of this software (version 2.1 beta 2) is included in this month's DVD of EFY Plus.

What is Essentia about

The software package is itself a collection of algorithms for feature extraction from audios, wrapped up as a C++ library. When it comes to analysis of audio data, ease of use, performance, maintainability of the code, accuracy, optimality and availability of algorithms are the major design considerations. Essentia is developed to cater to all these requirements.

Decision and implementation of

the flow of music analysis is completely left to the discretion of the user. The software simply takes care of implementation details of the algorithms used.

The reason behind development of Essentia as a C++ library is to ensure high performance. Besides this, the object-oriented programming environment in C++ is well-suited for audio-signal processing. Python bindings provided in the package make it well-suited to be used in an interactive development environment. Writing feature extractors is further made easy by the introduction of streaming mode, along with the conventional standard mode.

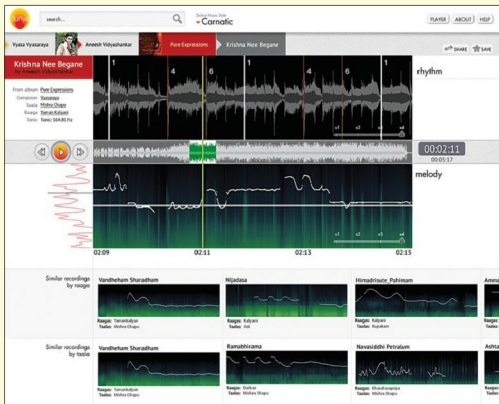
Further, conversion of Python code into C++ in the streaming mode is rather straightforward. It also

includes a Vamp plugin to be used with Sonic Visualiser for visualisation purposes.

Where can Essentia be used

The software has been used extensively in a wide range of applications in audio-signal processing. Essentia could be used for semantic autotagging, visualisation and interaction with music, sound indexing, cover and beat detection.

Researchers in neuroimaging make use of Essentia for the study of acoustic analysis of stimuli. It can also be used for classification of music, especially as per the mood, and for finding out similar music recommendations. We can use the software for recognising the instruments used in the music.



Visual display of a recording in Durya, a web based software application using Essentia